

Mitesh Kumar



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SUMMARY

Master's student in Data Science at FAU specializing in machine learning and mathematical optimization. Currently developing bicriteria optimization models for gas network systems at FAU, formulating novel objectives that balance economic efficiency with data-driven explainability. Experienced in computer vision and deep learning with proven results including 0.78 mAP on historical document retrieval and 90% accuracy in open-set face recognition. Seeking research opportunities in optimization, machine learning, or applied AI.

WORK EXPERIENCE

UNIVERSITY OF REGENSBURG

Student Research Assistant — Machine Learning for Side-Channel Analysis

Regensburg, Germany

12/2025 – Present

- Reviewing and synthesizing literature on deep learning-based side-channel attacks, including profiling and single-trace attacks.
- Analyzing ML-assisted key-recovery attacks on post-quantum signature schemes (e.g., multivariate cryptography).
- Investigating robustness of neural-network-based attacks under realistic measurement conditions (noise, misalignment).
- Supporting the design of experimental pipelines for power-trace acquisition and ML-based leakage analysis.

FRIEDRICH-ALEXANDER-UNIVERSITÄT ERLANGEN-NÜRNBERG

Student Research Assistant — Transregio 154: Mathematical modeling, simulation and optimization using the example of gas networks.

Erlangen, Germany

05/2025 – Present

- Engineered bicriteria optimization model in Pyomo balancing social welfare with explainability through dynamic pressure-based features and adaptive weighting (λ, β) .
- Scaled experiments across HPC cluster using Slurm, aggregating 50+ Pareto fronts via custom Min-Max normalization to resolve mathematical scaling distortions.
- Developed non-convex nominal model in Pyomo and Gurobi to maximize gas network social welfare subject to physical transport constraints.
- Generated large-scale time series datasets by simulating thousands of optimal network states through automated Python workflows.

SKILLS

Technical Skills

- Programming: Python, SQL, C++, LaTeX
- ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras
- Optimization: Pyomo, Gurobi
- Data & Visualization: Power BI, Pandas, NumPy, Matplotlib
- Tools: Git, Microsoft Office

Domain Expertise

- Machine Learning & Deep Learning
- Optimization & Mathematical Modeling
- Computer Vision
- Cryptographic Analysis

Language Expertise

- English (IELTS C1)
- Hindi (Native)
- German (Elementary)

PROJECTS

Hierarchical Object Detection via Selective Search & R-CNN Framework	01/2025 – 02/2026
- Engineered object detection pipeline integrating multi-scale Selective Search, ResNet18 CNN for feature extraction, and Linear SVM for classification. - Achieved MABO of 0.74 through Non-Maximum Suppression (NMS) and geometric data augmentation, demonstrating effective region proposal quality. - Developed region proposal algorithm using color, texture, size, and fill similarities on Felzenszwalb segmentation. - Trained RBF-Kernel SVM with balanced class weighting and automated dataset balancing.	
Open-Set Face Recognition & Robust Tracking System	12/2025 – 01/2026
- Achieved 90% validation accuracy in open-set identification by comparing Single and Multi Pseudo Label (SPL/MPL) strategies to distinguish known users from strangers. - Optimized efficiency via prediction pipeline using dynamic K-Means clustering for unknown identities and vectorized Nearest Neighbor searches. - Built face recognition pipeline using FaceNet (ResNet50) with dual-stream color/grayscale fusion for 128-dimensional embeddings. - Integrated MTCNN for detection with Template Matching fallback tracker for identity continuity during occlusions.	
Large-Scale Writer Retrieval via Multi-VLAD & Exemplar-SVM	11/2025 – 12/2025
- Achieved mAP of 0.78 on ICDAR17 dataset using Upright RootSIFT descriptors and Exemplar SVMs (One-vs-Rest classification). - Boosted baseline performance by 24% via Generalized Max Pooling (GMP) and PCA-whitened Multi-VLAD ensemble for discriminative representations. - Compressed local features into global representations using VLAD with power normalization and L2 scaling. - Integrated Generalized Max Pooling with balancing parameter (λ) to weight frequent versus rare visual words.	
Computational Imaging Pipeline: Demosaicing & HDR Reconstruction	10/2025 – 11/2025
- Synthesized HDR images from JPG sequences by reverse-engineering camera response function through mathematical optimization, recovering linear irradiance data. - Implemented custom pipeline from scratch including Bayer demosaicing and logarithmic tone-mapping to compress dynamic ranges without clipping artifacts. - Developed HDR reconstruction merging multiple exposures into radiance maps, preserving shadow and highlight detail. - Applied tone mapping via iCAM06 model and local contrast enhancement for HDR display.	
3D Scene Perception & Robust Box Dimension Estimation	09/2025 – 10/2025
- Extracted precise 3D dimensions from noisy ToF sensor data using MLESAC plane-fitting and morphological filters to eliminate background noise. - Optimized for real-time constraints via custom Preemptive RANSAC strategy, reducing computational overhead while maintaining geometric accuracy. - Implemented RANSAC variants (MLESAC, Preemptive RANSAC) for plane segmentation with sensor noise handling. - Automated object isolation via Connected Component Analysis, mapping 2D coordinates to 3D Euclidean space.	

EDUCATION

Universidad Politécnica de Madrid

Master of Science, Data Science (Erasmus Semester)

Madrid, Spain

02/2026 – Present

Friedrich-Alexander-Universität Erlangen-Nürnberg

Master of Science, Data Science (Grade: 2.0)

Erlangen, Germany

10/2024 – Present

Amity University

Bachelor of Science, Mathematics (Grade: 1.8)

Noida, India

07/2020 – 06/2023

ACHIEVEMENTS

- Presented Research Paper : Delivered a presentation on "*Modelling Intuitionistic Fuzzy Sets in Real Life Using a Multi-Criterion Decision-Making Approach*" at a National Conference organized by the Department of Mathematics, LNCT Group of Colleges, Bhopal, India
- Academic Excellence : Awarded a Bronze Medal for outstanding academic performance during Bachelor of Science (Mathematics) at Amity University, Noida, India